PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Codename: NIRMAAN / ChainTrace AI
SIH Problem Statement: SIH26183 — *Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics*
Issuing Authority: Ministry of Home Affairs (MHA) / Indian Cybercrime Coordination Centre (I4C)
Document Version: 1.0.0
Status: Approved for Scaffolding & Prototyping

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1. Executive Summary & Product Vision

1.1 Product Vision

To empower frontline Indian law enforcement officers (LEAs), cyber crime police stations (CCPS), and the national 1930/NCRP network with a **sovereign, automated, real-time blockchain tracing and VASP attribution engine** that intercepts illicit cryptocurrency flows during the critical **"Golden Hour"** and auto-generates court-admissible statutory notices under the **Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023**.

1.2 The Problem Statement

Indian citizens lost **₹22,845 Crore** to cyber fraud in 2024. Over **35% of high-value illicit proceeds** (task scams, digital arrest scams, fake trading apps) are rapidly layered through P2P transactions into **Tether (USDT)** on Tron (TRC-20) and EVM chains. While the 1930 / CFCFRMS system effectively freezes Layer-1/Layer-2 bank accounts, the trail goes dark once fiat converts to crypto.

Current foreign forensic solutions (Chainalysis Reactor, TRM Labs) cost **₹45–₹70 Lakhs/seat/year**, require weeks of manual training, and demand 30–60 minutes of manual graph expansion per case. NIRMAAN automates the entire investigative pipeline: **from victim wallet input to destination exchange identification and statutory Section 94 BNSS notice dispatch in under 10 seconds**.

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2. Target User Personas & User Journeys

USER PERSONA MATRIX		
Persona	Role & Environment	Primary Goal
Sub-Inspector (IO)	District Cyber Police Station; high case volume; zero Web3 skills.	Needs instant attributi suspect wallet & pre-fi
I4C Nodal Officer	MHA Central Operations; oversees 1930 & national suspect registry.	Automated syndicate clus cross-district linkage a
VASP Compliance Officer	Indian & offshore exchanges (CoinDCX, Binance, WazirX).	Automated ingestion of statutory preservation r
Bank Cyber Risk Manager	Commercial Banks (SBI, HDFC); monitors high-velocity UPI exits.	Early warning on P2P mu before police lien freez

2.1 Primary User Journey: The 60-Second Field Investigation

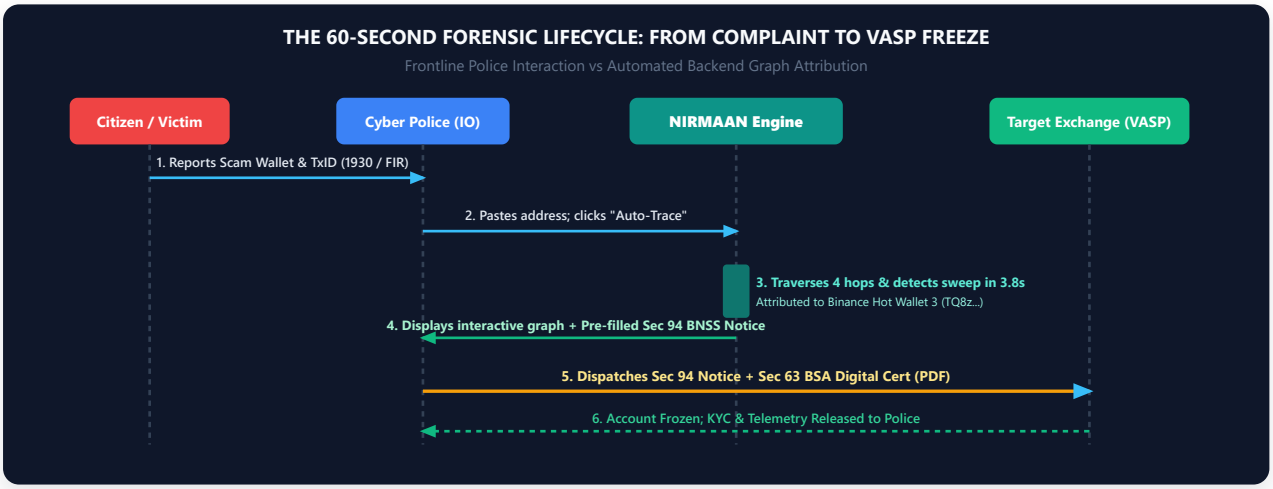


Figure 1.1: The 60-Second Field Investigation Lifecycle: From Citizen Complaint to VASP Asset Freeze

3. Product Scope & Functional Requirements

3.1 Functional Requirements Matrix

ID	MODULE	FEATURE DESCRIPTION	PRIORITY
FR-01	Multi-Chain Ingestion	Auto-detect address format (Tron Base58 `T...`, EVM Hex `0x...`, Bitcoin Bech32/Base58 `bc1...` / `1...`) and route to correct crawler.	**P0 (Must Have)**
FR-02	Automated BFS Traversal	Multi-hop transaction crawler traversing up to 5 hops with dynamic value pruning ($> \$20$ USD threshold) and timestamp clipping.	**P0 (Must Have)**
FR-03	Deposit Sweep Heuristic	Deterministic identification of unlabelled user deposit addresses via parent gas-funding + sweep pattern on EVM & Tron Stake 2.0 energy delegations.	**P0 (Must Have)**
FR-04	VASP Directory & Matching	In-memory lookup matching resolved nodes against curated directory of 500+ global & Indian FIU-registered exchange hot wallets.	**P0 (Must Have)**
FR-05	Interactive Visualizer	Interactive Cytoscape.js canvas rendering: (1) Dagre linear fund flow view, (2) CoSE entity clustering view, with color-coded nodes.	**P0 (Must Have)**
FR-06	Legal Dossier Generator	One-click export of court-admissible PDF dossier containing transaction ledger, graph snapshot, Section 94 BNSS notice, and Section 63 BSA certificate.	**P0 (Must Have)**
FR-07	Peeling Chain Detector	Automated identification of UTXO / account-based peel chains (asymmetric 90/10 splits rolling into virgin change addresses).	**P1 (Should Have)**
FR-08	Mixer / Bridge Detection	Flags interactions with Tornado Cash, FixedFloat, ChangeNOW, Thorchain, and known cross-chain bridges.	**P1 (Should Have)**
FR-09	Case Management Vault	SQLite / Postgres case repository tracking FIR number, IO details, victim loss, notes, and timestamped audit logs.	**P1 (Should Have)**
FR-10	Dynamic API Rate Limiter	Redis-backed token bucket preventing HTTP 429 throttling across TronGrid and Etherscan APIs.	**P1 (Should Have)**

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4. Non-Functional Requirements (NFRs)

- **Performance & Latency:**
- Multi-hop trace latency $\le 5.0\text{ seconds}$ for up to 4 hops on Tron TRC-20 and EVM chains.

- Graph rendering time $\le 500\text{ ms}$ for graphs containing up to 2,500 nodes.
- **Accuracy & Attribution Confidence:**
- False positive rate for VASP attribution $\le 1.0\%$.
- Deposit sweep heuristics must achieve $\ge 98.0\%$ precision by validating parent gas/energy linkages.
- **Security & Sovereign Compliance:**
- Zero citizen PII transmitted to public blockchain explorers.
- System architecture must be deployable on MeitY-empanelled cloud or on-premise State Data Centres (SDCs).
- **Legal Admissibility:**
- All generated forensic reports must produce a **cryptographic SHA-256 hash** of raw transaction ledgers, fulfilling Section 63(4)(c) of the Bharatiya Sakshya Adhiniyam, 2023.

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5. Success Metrics & Key Performance Indicators (KPIs)

1. **Mean Time to Attribution (MTTA):** Reduced from **14 days** (manual/legacy) to **$< 10\text{ seconds}$**.
2. **Investigation Cost per Station:** Reduced from **₹50 Lakhs/seat/year** (Chainalysis) to **$< ₹3 Lakhs/year** per cyber station bundle.
3. **Legal Notice Turnaround Time:** Decreased from **72 hours** of manual paperwork to **under 2 minutes**.
4. **VASP Trace Completion Rate:** Successful resolution of $> 85\%$ of victim-reported USDT scam flows into identifiable exchange endpoints.